

Total No. of Questions : 6]
P4881

SEAT No. :
[Total No. of Pages : 2

TE/Insem. - 147
T.E. (Computer Engineering)
OPERATING SYSTEMS DESIGN
(2012 Pattern) (Semester - I)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates :

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.

- Q1)** a) What do you understand by free space management? Explain its different approaches. [6]
b) Explain booting process in detail. [4]

OR

- Q2)** a) Explain algorithm bread() and bwrite() [6]
b) Explain with neat diagram buffer header. [4]
- Q3)** a) Explain with diagram saving context of process. [4]
b) Compare and contrast various approaches for Deadlock handling. [4]
c) What is thread? [2]

OR

- Q4)** a) What is a process? Explain five state process model with diagram. [5]
b) With given matrices, determine safe state with the help of banker algorithm. [5]

Claim Matrix C	Allocation Matrix A	Resource Vector R
R1 R2 R3	R1 R2 R3	R1 R2 R3
P1 3 2 2	P1 1 0 0	9 3 6
P2 6 1 3	P2 6 1 2	
P3 3 1 4	P3 2 1 1	
P4 4 2 2	P4 0 0 2	

Available Vector V: R1-0 R2-1, R3-1

PTO.

- Q5)** a) Write short note on Thrashing. [2]
- b) Explain with example any two page replacement algorithms – FIFO, Optimal, LRU. Page address stream {2,3,2,1,5,2,4,5,3,2,5,2}, frame size –3. Identify the page faults occurred. [8]

OR

- Q6)** a) Why is the principle of locality crucial to the use of virtual memory? Explain with example. [4]
- b) What is TLB? Explain operation of TLB with neat diagram. [6]

